Comprehensive Career Development and Recruitment Framework using Multi-Causal AI

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Project Proposal Report

Kaluarachchi S.M – IT22112446

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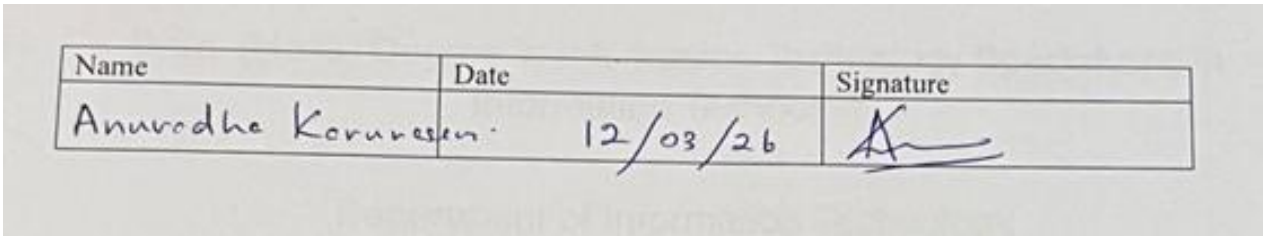
DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
IT22112446	Kaluarachchi S.M	

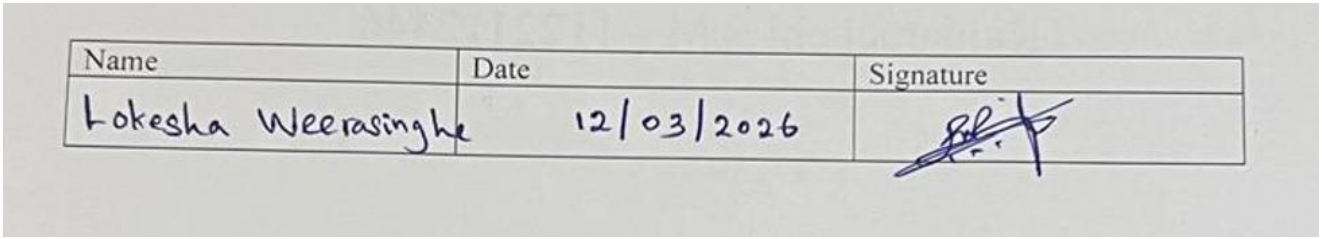
The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Supervisor



Name	Date	Signature
Anuradha Karunaratne	12/03/26	

Co-Supervisor



Name	Date	Signature
Lokesha Weerasinghe	12/03/2026	

ABSTRACT

Due to the rapid development in technology and the changing demands in the labor market, there has been a significant gap between formal education and preparation for employment. There are issues that students and employment seekers face in determining their levels of interview readiness, communication skills, and suitability for certain career paths. Today's systems for providing career guidance and hiring are based on resume-based data, psychological tests, and artificial intelligence methods, which are often difficult to understand. Therefore, these systems are often geared towards providing probabilities for hiring rather than assisting users in becoming more career ready.

This study aims to propose an "AI-Driven Interview Readiness and Career Path Guidance System" that will allow users to have better understanding and knowledge about their levels of communication readiness and suitability for certain careers by using non-intrusive and ethical means of data analysis. The proposed system will incorporate structured data from resumes, LinkedIn, labor market trends, and interview transcript analysis to determine the candidate's levels of communication readiness and representation of skills. In contrast to today's systems that are often based on artificial intelligence methods, the proposed system will allow users to have a better understanding of their levels of career readiness through qualitative analysis.

Explainable machine learning methods, including decision tree analysis, will be used to determine the results. In addition, the proposed system will allow users to conduct what-if analysis to determine how improvements in communication or skill representation can affect their levels of career readiness.

The proposed AI-driven decision support system will allow users to become more effective in preparing for interviews and matching their professional profiles with realistic career paths. The proposed solution will aim to improve levels of employability readiness, promote the responsible use of AI in developing career-oriented technologies, and promote transparency in hiring systems.

Keywords: Explainable AI, Interview Readiness, Career Guidance, Employability Intelligence, Ethical AI

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LIST OF ABBREVIATIONS

Table 1: List of all abbreviations used in this report

Abbreviation	Description
AI	Artificial Intelligence
CV	Curriculum Vitae
DSR	Design Science Research
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
SDG	Sustainable Development Goal
UI	User Interface
UX	User Experience
CSV	Comma Separated Values
API	Application Programming Interface
IT	Information Technology
WEF	World Economic Forum
IEEE	Institute of Electrical and Electronics Engineers
SLIIT	Sri Lanka Institute of Information Technology

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AI-Driven Interview Readiness and Career Path Guidance System

1.INTRODUCTION

1.1 Background and Context

Significant changes in the global employment landscape have been brought about by the rapid developments in technology and the ever-changing requirements of the labor market. As industries increasingly require employees to have high levels of digital competence, effective communication skills, and flexible professional skills, there has been an ever-widening gap between formal education and the actual state of employability. Many students face problems in determining their levels of interview preparation, developing effective communication skills, and finding careers that are realistically compatible with their skills and professional interests.

As per the World Economic Forum's "Future of Jobs Report 2023," it has been estimated that 44% of the main skills of workers are expected to change by 2027 due to technological transformation, automation, and innovation. Therefore, it has become increasingly important to have effective systems for providing guidance to employees in determining their levels of preparation for interviews, developing their communication skills, and becoming compatible with the changing requirements of the job market. In addition, many IT professionals, as per the Stack Overflow Developer Survey, have cited that developing effective communication skills and problem-solving skills are essential for advancing their careers; however, many graduates feel that they are not equipped to effectively communicate their skills during interviews.

The current career guidance and recruitment practices mainly depend on resumes, academic qualifications, psychometric tests, and keyword-based filters. Online career platforms provide career recommendations based on candidates' interests, whereas recruitment platforms use automated shortlisting tools to filter out suitable candidates. These practices often heavily depend on static information and lack sufficient information about the ability of candidates to effectively communicate their knowledge, skills, and realistic career paths. Moreover, current recruitment practices heavily use sophisticated artificial intelligence models, which are essentially "black box" models. This makes users unable to comprehend the results of the evaluations carried out by these models. These models often provide users with scores or probabilities instead of meaningful results that can help users improve their interview skills.

Another significant concern relates to the ethical and transparency limitations associated with some of the recruitment technologies that are based on AI. Some of these technologies are based on data collection, personality, and decision-making processes that are likely to undermine user trust and raise ethical concerns. As such, there is an increased need for ethical and transparent AI technologies that are geared towards career development and interview preparation without relying on evaluation mechanisms and decision-making processes.

Therefore, in addressing the challenges that are associated with the current state of the art in recruitment technologies, this research proposes an AI-Driven Interview Readiness and Career Path Guidance System that uses explainable machine learning approaches to evaluate interview communication indicators and professional background data. Unlike traditional interview preparation and recruitment technologies, the proposed system does not use numerical scoring and decision-making processes. Instead, it uses an explanation-based approach that enables users to understand their strengths and possible career paths based on their communication and skill abilities.

The proposed research would also contribute to the achievement of various United Nations Sustainable Development Goals, i.e., the United Nations Sustainable Development Goals 4 - Quality Education, focusing on the readiness for lifelong learning and employability, and the United Nations Sustainable Development Goals 8 - Decent Work and Economic Growth, focusing on the sustainability of employment opportunities and the alignment of the workforce to the needs of the industry. This would enable individuals to make the right choices for their careers, as well as improve the productivity and job satisfaction of the workforce.

Furthermore, the proposed research would also align with the Artificial Intelligence and Data Science Research Cluster of the Sri Lanka Institute of Information Technology Research Framework. This would focus on the importance of the development of analytical models, natural language processing, and the use of data-driven solutions for the improvement of career development. This would enable the use of ethical artificial intelligence solutions for the improvement of career intelligence systems.

1.2 Literature Review

Artificial Intelligence (AI) has also found its place in the field of recruitment, career development, and professional guidance systems. AI methodologies allow for the automatic analysis of user skills, educational background, and professional interests to provide relevant information for a better career path. With the growth of digital job platforms and online learning environments, AI-based systems are being used to help individuals identify their respective career paths and improve their preparedness for available job opportunities.

Existing Research Approaches:

There are several researchers who have attempted to utilize AI and machine learning methodologies to improve career guidance systems. AI-based mock interview systems are being used to mimic real interview environments with automated feedback systems to improve candidate performance. For example, Chavan and Jadhav have suggested an AI-based mock interview system to improve candidate performance with relevant feedback to improve communication skills [1]. Such systems allow candidates to interact in a simulated interview environment to improve their responses to interviewers.

Similarly, AI-based career guidance systems have been developed that analyze user profiles, skills, and interests to provide appropriate recommendations. N. B. B. has proposed an AI-based career guidance system that assists users in finding appropriate careers based on their skills and interests

[2]. Other researchers have focused on developing AI-based career recommendation systems that analyze large datasets containing skills, labor market trends, and educational qualifications to provide appropriate recommendations for users [3][4].

Machine learning algorithms and recommendation models have been utilized in the development of AI-based career decision-making systems. For example, Wadkar et al. have proposed an AI-based career path advisor that uses skill analysis and recommendation techniques to provide appropriate recommendations for users [5]. These systems utilize prediction models and data analytics techniques to identify appropriate career opportunities based on user skills and industry demands.

Another interesting research area is the development of intelligent resume analysis and feedback systems. Mishra et al. have proposed an AI-based resume recommendation model that analyzes resumes and provides improvement suggestions based on job requirements and skills [6]. Additionally, resume analyzer systems have been proposed in the literature to analyze candidate resumes and provide appropriate recommendations for interview preparation and skills development [7].

Existing Products and Services:

Apart from academic research, several products and services, both for business and research purposes, are already providing career guidance services with the help of AI. These include career guidance tools that use AI to provide personalized career guidance to users. These tools use various parameters such as the profile, resume, and interests of the users to provide guidance. An example of this is the use of AI algorithms for providing career guidance. These algorithms use machine learning to evaluate the skills of the users and suggest an appropriate career option for them. These are a few examples of the use of AI for providing career guidance to users.

In addition to this, several integrated products are also available that provide resume analysis, interview preparation, and career guidance to users. Shenoy et al. proposed a product known as Career Link, which uses AI for resume analysis, interview preparation, and career guidance to enhance the employability of users.

In addition to this, several career guidance tools are available that use AI algorithms to provide career guidance to users. These tools are particularly helpful for students and new professionals. These tools use various parameters such as academic performance, skill levels, and interests to provide guidance to the users. These tools are helpful for users who are not aware of their career paths. An example of such a tool is provided by Thakur et al. These tools use AI algorithms to provide guidance to the users based on their academic performance, skill levels, and interests.

Emerging AI Technologies in Career Guidance:

Recent developments in AI technology, particularly in natural language processing (NLP) and large language models (LLMs), have also contributed to the development of more effective career guidance systems. For example, NLP technology has shown promise in analyzing texts such as resumes, personal statements, and career interests to forecast an individual's potential career paths. For instance, Faruque et al. presented a natural language processing-based framework for career prediction using textual data analysis to identify potential career paths for an individual [11].

Furthermore, conversational AI technology has also enabled the development of interactive career guidance systems. For example, Daryanto et al. presented an AI-based interview simulation system that allows an individual to simulate an interview process while providing interactive learning support [12]. Moreover, chatbots using AI technology in the form of LLMs can also be used to provide guidance to an individual in their career development journey through interactive conversations [13].

Additionally, simulation AI technology has also shown promise in allowing an individual to explore various potential career paths in a virtual environment. For example, Wang et al. presented an AI-based simulation system known as CareerPooler that allows an individual to explore various potential career paths and gain a deeper understanding of industry needs [14].

Limitations of Existing Systems:

Despite the significant progress in the development of AI-driven career guidance technologies, some limitations still need to be addressed. Some of the limitations include the fact that most of the available technologies are designed to perform particular tasks, such as resume analysis and job recommendations, without offering an integrated system that combines career guidance, interview readiness, and skill development.

Another limitation that needs to be addressed in the development of AI-driven career guidance technologies is the lack of transparency and explainability in most of the available systems. The literature suggests that some of the available technologies, particularly those that utilize AI, often offer limited explanations on how they generate their recommendations.

Another limitation that needs to be addressed in the development of AI-driven career guidance technologies is the fact that some of the available technologies often collect significant amounts of personal data, thus raising concerns about data privacy. As such, there is an increased need for developing AI technologies that utilize non-intrusive data sources, such as resumes, skill sets, and academic records, in offering accurate and meaningful career recommendations.

From the literature, it is evident that AI-driven career guidance technologies have significant potential in assisting individuals in their career paths and interview readiness. However, there is still an increased need for developing integrated technologies that combine personalized career

recommendations, interview readiness, and data analysis, thus leading to the development of the proposed AI-Driven Interview Readiness and Career Path Guidance System.

1.3 Research Gap

Even though there are numerous AI-based tools for recruitment and career recommendations, there is a lack of an integrated system that is able to provide communication analysis in interviews with appropriate recommendations for the best career path based on non-intrusive data. Most of the existing tools provide recommendations based on static data such as resumes and academic records, which do not show the ability of the candidates to communicate skills, knowledge, and professional competencies in interviews.

The existing tools are not effective in the current recruitment process as they provide only numerical or categorical recommendations without appropriate feedback, are opaque in nature, and sometimes require intrusive data collection methods such as behavioral or biological monitoring. Therefore, the existing tools are not appropriate for the recruitment process in terms of user trust, transparency, and ethical concerns. The proposed research is an attempt to bridge the gap in the existing tools by developing an AI-based system that is able to provide appropriate recommendations for interview performance with qualitative recommendations for the best career path. The novelty of the proposed research is the integration of the interview performance assessment with appropriate recommendations for the best career path in a transparent, ethical, and non-intrusive manner, which is not available in the existing tools.

The need for the proposed research is evident from the fact that the existing AI tools based on the black box algorithm have been shown to reduce user trust, whereas the existing scoring tools are not able to provide appropriate recommendations for improvement in the performance of the candidates in interviews. The recent findings by LinkedIn and the World Economic Forum indicate the need for developing appropriate AI tools for career recommendations in a transparent, ethical, and non-intrusive manner. Therefore, there is a research gap in the existing tools in terms of the integration of communication assessment with appropriate recommendations for the best career path in a user-friendly manner.

Table 2:Research gap analysis

Research Area / Existing Approach	**What Previous Systems Do**	**Limitations Identified in Literature**	**Research Gap**	**Proposed System Contribution**
Resume-Based Recruitment Systems	Many AI recruitment tools analyze resumes, academic records, and skill keywords to screen or rank candidates.	Evaluation depends on static data sources, which do not represent how candidates communicate their knowledge or professional competencies during interviews.	Lack of mechanisms to analyze real interview communication behavior for career readiness assessment.	The proposed system analyzes interview response patterns using NLP to understand how candidates express their skills and competencies.
AI-Based Career Recommendation Platforms	Career platforms recommend career paths using skill matching, personality tests, or historical job data**.**	These systems do not evaluate interview readiness or communication ability, which are important for career success.	Missing integration between interview performance analysis and career guidance**.**	The proposed system integrates interview communication insights with career path recommendations.
Automated Candidate Scoring Models	Machine learning models generate candidate scores, rankings, or predictions for recruitment decisions.	Systems often produce numeric scores or binary classifications without actionable feedback for improvement.	Lack of qualitative explanations and improvement guidance for candidates.	The proposed system provides explainable feedback and qualitative guidance instead of simple scores.
Black-Box AI Recruitment Tools	Advanced AI models such as deep learning are used for automated decision making.	Many models operate as black-box systems, reducing transparency and user trust.	Limited interpretability and transparency in recruitment-support AI systems.	The proposed system uses Explainable AI (XAI) to provide transparent reasoning behind recommendations.
Data-Intensive or Intrusive Assessment Systems	Some modern systems use behavioral monitoring, facial recognition, or biometric analysis during interviews.	These approaches raise privacy concerns, ethical issues, and user discomfort.	Need for ethical, privacy-preserving, and non-intrusive evaluation approaches**.**	The proposed system uses non-intrusive textual data such as interview transcripts, CVs, and job descriptions.
Integration of Interview	Interview preparation	Lack of an integrated system	Absence of a unified	The proposed system introduces

Readiness and Career Development	platforms and career guidance tools usually exist as separate solutions.	that supports interview readiness evaluation together with career decision support.	explainable framework connecting interview communication performance with career guidance.	an integrated explainable AI framework combining interview readiness analysis with career path guidance.

As shown in the table, existing recruitment and career guidance systems mainly focus on resume analysis, candidate scoring, or personality-based recommendations. However, they do not analyze interview communication patterns, often operate as black-box systems, and frequently provide only numeric predictions without actionable guidance. Additionally, some approaches rely on intrusive data collection methods. Therefore, this research addresses these limitations by proposing an explainable, non-intrusive AI system that evaluates interview readiness and integrates communication insights with personalized career path guidance

2.OBJECTIVE

2.1 Main Objective

To develop an AI-driven interview readiness and career path guidance system that evaluates candidates' communication effectiveness during interviews and aligns them with suitable career opportunities using explainable AI models and non-intrusive data sources.

2.2 Specific Objectives

1. Ethically obtaining professional information from resumes, LinkedIn, and other relevant job market sites while maintaining privacy.
2. Obtaining indicators for interaction with the interviewer, such as communication clarity, structuring of answers, and skill explanation, from interviews or simulated interviews.
3. Analyzing the effectiveness of communication and career alignment for candidates using explainable machine learning to provide transparent results.
4. Providing qualitative results for interview preparedness rather than solely numerical or classified results.
5. Creating scenario-based "what-if" results to assist the candidate in improving their performance for the interview and career alignment.
6. Creating a system that incorporates all these elements to provide a user-friendly interface for career guidance while maintaining ethics and privacy.

3.METHODOLOGY

This research utilizes the Design Science Research (DSR) approach along with the experimental protocol. DSR is applicable to this research since the aim is to design, implement, and evaluate the AI-based system that enables the individual to be more prepared for interviews and provides guidance on their career paths with the help of non-intrusive approaches.

There are two primary phases of the methodology:

Phase 1: System Design and Development

1.Data Integration:

- Data sources include curricula vitae, LinkedIn, and interviews.
- Data collection is limited to ethically obtained non-intrusive data to maintain privacy.

2.Feature Extraction:

Natural Language Processing (NLP) is used to identify communication features such as:

- Clarity of answers
- Structural coherence
- Articulation of skills
- Confidence and professional tone

3.Explainable Machine Learning Models:

- Data analysis is performed using Decision Trees.
- Model selection for this system prioritizes interpretability to allow users to understand the rationale behind the suggestions.

4.Qualitative Output Generation:

The system offers suggestions that highlight:

• Communication strengths

• Improvements to be made

• Suggestions for career alignment

5.Scenario-Based Simulations:

- Scenario analysis shows the effect on interviews and career suitability based on changes to communication style and skill articulation.

Phase 2: Experimental Evaluation

1.Validation Strategy:

- Cross-validation ensures model consistency and stability across different inputs.
- Evaluation emphasizes the accuracy and interpretability of qualitative guidance rather than numeric scoring.

2.User Feedback Evaluation:

- Candidate and expert feedback are collected via surveys and interviews.
- Measures include:
  - Clarity and usefulness of guidance
  - User understanding and perceived improvement
  - Alignment with career goals

3.Reliability Checks:

- NLP features and extracted communication indicators are validated for consistency and dependability.
- The system is tested to ensure stable recommendations across similar interview scenarios.

3.1 Technical Methods and Algorithms

- NLP Feature Extraction: Analyzes interview transcripts to quantify communication quality.
- Explainable ML Models: Decision Trees and rule-based algorithms for transparent analysis.
- Scenario-Based Guidance: Simulates improvements in communication or skill articulation to guide career alignment.

3.2 Tools and Technologies

Table 3:Tools and Technologies

Tool / Platform	Purpose	Justification
Google Forms	Data collection	Free, mobile-friendly, supports CSV export, easy for participants, ethical.
Python	Implementation	Strong support for NLP, ML, and reproducible workflows; flexible for research prototyping.
Pandas / NumPy	Data handling	Efficient preprocessing, feature extraction, and structured dataset preparation.
Scikit-learn	ML training & evaluation	Ready-to-use interpretable models, cross-validation, and reproducible pipelines.
Jupyter Notebook / VS Code	Development environment	Supports exploratory analysis (Jupyter) and structured coding (VS Code).

3.3 Data Collection and Ethical Considerations

Target Population:

- Final-year undergraduate [1]s, recent graduates (0–2 years), and early-career professionals (0–3 years' experience) in Sri Lanka.

Sampling Method:

- Participants recruited via university career centers, LinkedIn graduate groups, and professional networks.

Ethical Considerations:

- Only non-intrusive, professional data is collected.
- Information consent is obtained from all participants.
- No sensitive or biometric data is used, ensuring privacy and ethical compliance.

3.4 Validation Strategy and Performance Metrics

- Model Evaluation:
  - Focuses on the quality and interpretability of qualitative recommendations.
  - Cross-validation ensures consistent outputs from the ML models.

- User Feedback Validation:
    - Candidates and experts provide qualitative feedback on:
        - Clarity of guidance
        - Usefulness for interview readiness
        - Alignment with career goals

- Reliability Checks:
    - NLP features and communication indicators are tested for internal consistency.
    - The system is verified to produce stable and actionable recommendations across similar input scenarios.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

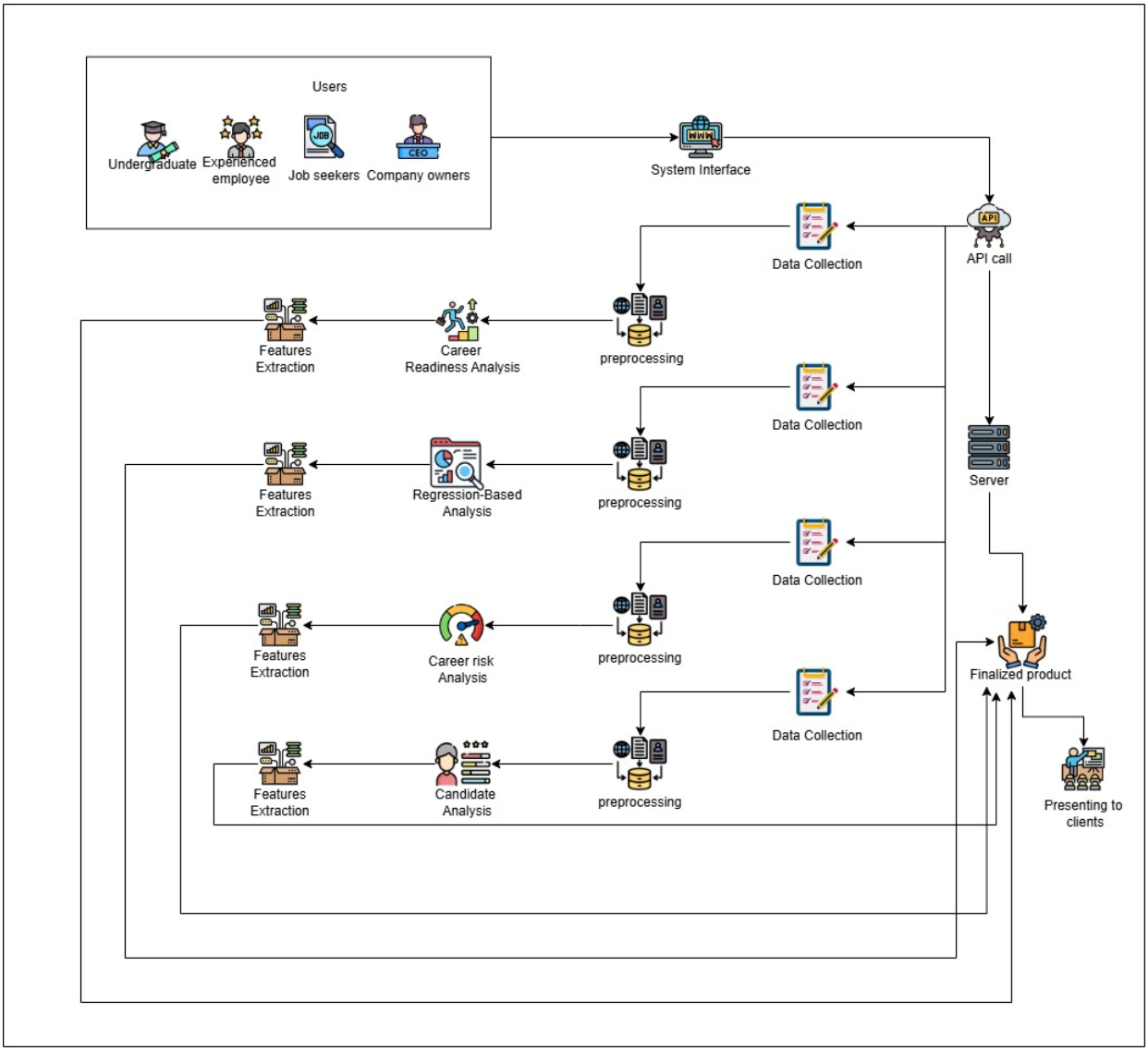


Figure 1:High-Level System Architecture Diagram

5.USER REQUIREMENTS

5.1 Stakeholders

The system involves multiple stakeholders who will benefit from AI-driven interview readiness and career guidance:

- Students – University students preparing for internships or job interviews.
- Graduates – Recent graduates seeking employment opportunities and career alignment.
- Job Seekers – Individuals looking to improve interview skills and explore suitable career paths.
- Career Advisors – Professionals providing guidance and feedback on career planning.
- Employers – Organizations interested in assessing candidate preparedness for interviews.

5.2 Functional Requirements

The system is designed to provide the following functionalities:

Table 4:Functional Requirements of the Proposed System

ID	Requirement Description
FR1	Users should be able to upload their CV for analysis.
FR2	Users should be able to connect their LinkedIn profiles for professional data integration.
FR3	The system should analyze interview responses using NLP techniques to extract communication indicators.
FR4	The system should generate qualitative interview readiness insights highlighting strengths and areas for improvement.
FR5	The system should provide personalized career path recommendations based on skills, experience, and interview performance.
FR6	The system should provide actionable suggestions to improve communication and professional presentation.

5.3 Non-Functional Requirements

The system will meet the following non-functional standards:

Table 5:Non-Functional Requirements of the Proposed System

Requirement	**Description**
Security	Ensure safe and secure storage of all user data, with access control mechanisms.
Privacy	Protect user information and ensure that all data usage is based on informed consent**.**
Usability	Provide an intuitive and user-friendly interface that is easy to navigate for all users.
Performance	Generate interview readiness insights and career recommendations efficiently without delays.
Explainability	Ensure AI recommendations are transparent and interpretable, allowing users to understand why a suggestion or insight is provided.

6.COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the AI-driven Interview Readiness and Career Path Guidance System includes:

- Students: University students preparing for internships or first-time job interviews.
- Graduates: Recent graduates seeking career alignment and professional growth opportunities.
- Job Seekers: Individuals aiming to improve interview performance and explore suitable career paths.
- Career Advisors and Coaches: Professionals supporting clients in career planning and skill development.
- Employers/Recruiters: Organizations interested in candidate readiness evaluation to streamline recruitment.

Customer Persona Example:

- Name: Ravi Perera
- Age: 23
- Profile: Final-year undergraduate preparing for job interviews in IT and business sectors.
- Needs: Personalized feedback on interview communication, guidance on career paths, and practical improvement strategies.
- Pain Points: Limited access to transparent AI-driven feedback, unclear career options, lack of actionable interview guidance.

6.2 Value Proposition

The system provides personalized, actionable, and explainable guidance to improve interview performance and align career choices with individual strengths. Key benefits include:

- Non-intrusive assessment: Uses professional data without collecting sensitive or biometric information.
- Explainable AI: Users can understand why recommendations are given, increasing trust.
- Career alignment: Matches communication strengths with suitable career opportunities.

- Scenario-based learning: Simulates “what-if” scenarios to show how improvements affect outcomes.

6.3 Revenue Model

Potential revenue streams include:

- Subscription Model: Monthly or annual subscription for students, graduates, and job seekers.
- Institutional Licenses: Universities, career centers, and coaching institutes can purchase site-wide licenses.
- Freemium Model: Basic guidance for free, premium features for detailed insights and scenario simulations.

6.4 Pricing Strategy

- Individual Users: Free basic version; premium subscription at LKR 1,500/month or LKR 15,000/year.
- Institutions**:** Tiered licensing based on number of users, starting at LKR 50,000/year for small-scale use.
- Workshops & Training Modules: Optional paid sessions for students or graduates at LKR 2,000–5,000 per participant.

6.5 Competitive Advantage

- Explainable AI: Unlike black-box systems, the model provides transparent reasoning behind recommendations.
- Non-intrusive**:** Protects privacy by avoiding sensitive biometric or behavioral monitoring.
- Integrated Career Guidance: Combines interview readiness with career path recommendations, unlike platforms that only focus on resume scoring.
- Scenario-Based Guidance: Provides actionable suggestions, helping users actively improve interview performance.

6.6 Intellectual Property Considerations

- AI models, feature extraction algorithms, and system workflow will be copyrighted and documented.
- Open-source libraries and tools will comply with licensing regulations.
- Proprietary data preprocessing techniques, scenario simulation logic, and qualitative guidance modules may be patented or protected under institutional IP guidelines.

7.BUDGET AND JUSTIFICATION

Table 6:Budget Estimation

Description	Amount (LKR)	Justification
Deployment and hosting	20,000	For cloud hosting and server infrastructure to run the system reliably.
Cloud services	5,000	Cloud computing for NLP processing and data storage.
Training and Databases	40,000	For AI model training and access to professional datasets
Travelling	4,000	For field visits, data collection, and stakeholder interviews.

8.WORK BREAKDOWN STRUCTURE (WBS)

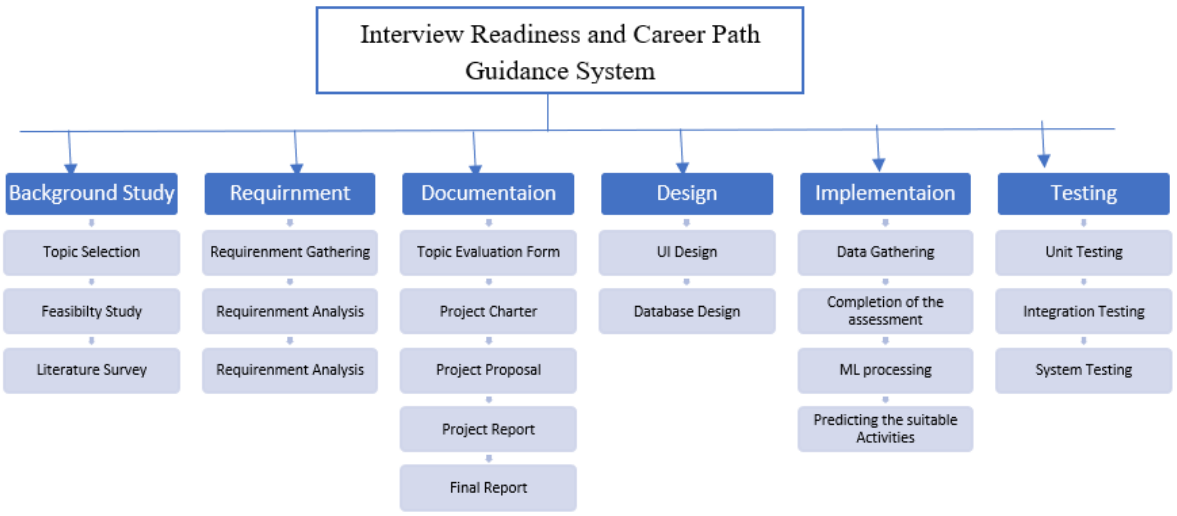


Figure 2:Work Breakdown Structure (WBS) Diagram

9.GANTT CHART

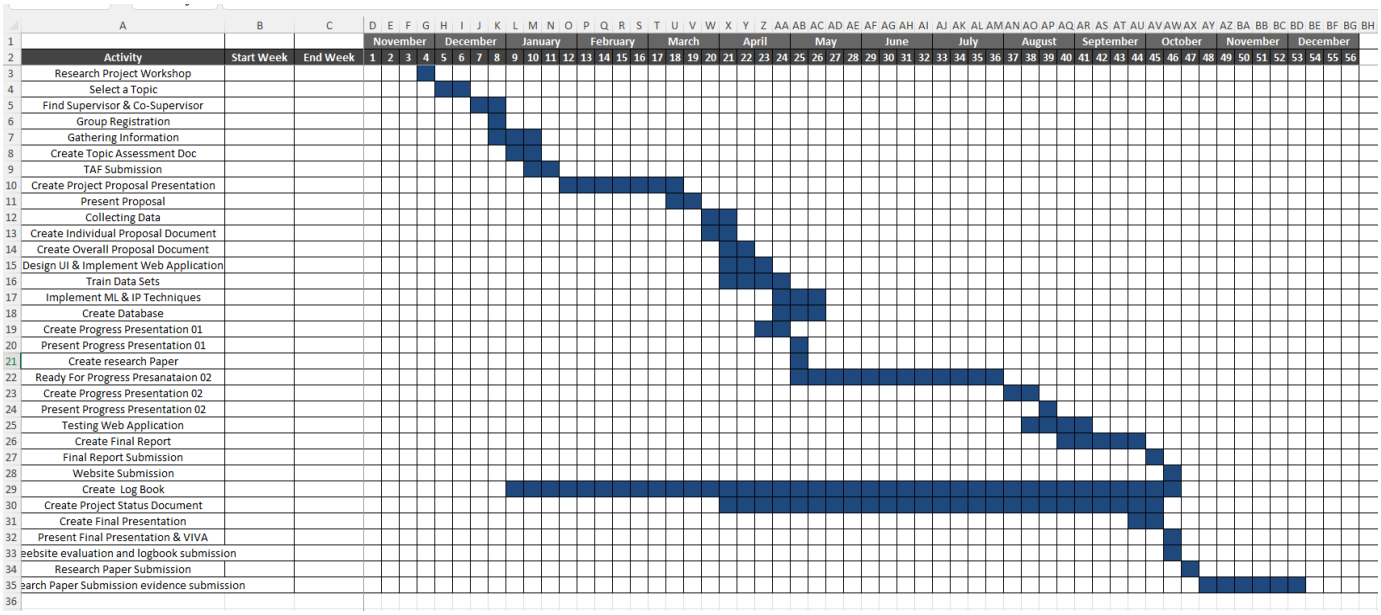


Figure 3:Project Gantt Chart

10.REFERENCES LIST

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11.APPENDICES

11.1 Survey instruments

	Present study	Miller and colleagues (2009)
Self-appraisal		
Accurately assess your abilities (5)	.62	.63
Determine what your ideal job would be (9)	.70	.76
Decide what you value most in an occupation (14)	.63	.64
Figure out what you are and are not ready to sacrifice to achieve your career goals (18)	.54	.73
Define the type of lifestyle you would like to live (22)	.61	.61
Occupational information		
Find information in the library about occupations you are interested in (1)	.48	.50
Find out the employment trends for an occupation over the next 10 years (10)	.59	.64
Find out about the average yearly earnings of people in an occupation (15)	.49	.63
Talk with a person already employed in the field you are interested in (19)	.60	.69
Find information about graduate or professional schools (23)	.65	.70
Goal selection		
Select one major (*field of study*) from a list of potential majors (*study courses*) you are considering (2)	.68	.74
Select one occupation from a list of potential occupations you are considering (6)	.71	.75
Choose a career that will fit your preferred lifestyle (11)	.63	.74
Make a career decision and then not worry about whether it was right or wrong (16)	.35	.49
Choose a major (*field of study*) or career that will fit your interests (20)	.70	.75
Planning		
Make a plan of your goals for the next 5 years (3)	.63	.61
Determine the steps you need to take to successfully complete your chosen major (*study course*; 7)	.66	.75
Prepare a good resume (12)	.54	.49
Identify employers, firms, and institutions relevant to your career possibilities (21)	.64	.73
Successfully manage the job interview process (24)	.61	.57
Problem solving		
Determine the steps to take if you are having academic trouble with an aspect of your chosen major (*sector of study*; 4)	.62	.63
Persistently work at your major (*scholastic*) or career goal even when you get frustrated (8)	.60	.65
Change majors (*field of study*) if you did not like your first choice (13)	.58	.57
Change occupations if you are not satisfied with the one you enter (17)	.63	.53
Identify from reasonable major (*field of study*) or career alternatives if you are unable to get your first choice (25)	.64	.72

Note. In italic the Italian adaptation.

Appendix A 1:Survey Instrument - Participant Consent Form

Survey Sections & Questions (40 Questions)

Section A: Demographics (5 Questions)

1. Age group: 18-22 / 23-27 / 28-32 / 33+
2. Current status: Student / Graduate / Employed / Unemployed
3. Field of study/work: Technology / Business / Engineering / Other
4. Years of work experience: 0 / 1-2 / 3-5 / 5+
5. Location: Urban / Suburban / Rural

Section B: Interview Experience (10 Questions)

6. How many job interviews have you attended in the past year?
7. Rate your confidence level during interviews (1-5 scale)
8. What aspects of interviews do you find most challenging? (Multiple choice)
9. Have you used any interview preparation tools? (Yes/No)
10. If yes, which tools? (Open-ended)
11. How do you typically prepare for interviews? (Multiple choice)
12. Rate the effectiveness of your current preparation methods (1-5)
13. What interview skills do you need to improve most? (Rank top 3)
14. Have you received professional interview coaching? (Yes/No)
15. How important is personalized feedback for interview improvement? (1-5)

Section C: Career Guidance Needs (10 Questions)

16. How clear are you about your career path? (1-5 scale)
17. Have you used career guidance services? (Yes/No)
18. What career information do you need most? (Multiple choice)
19. How do you currently explore career options? (Multiple choice)
20. Rate your satisfaction with available career guidance resources (1-5)
21. What factors influence your career decisions most? (Rank top 3)
22. Do you feel your skills match your desired career? (Yes/No/Unsure)
23. How important is understanding industry requirements? (1-5)
24. Would you trust AI-based career recommendations? (1-5)
25. What makes career guidance valuable to you? (Open-ended)

Section D: AI System Expectations (10 Questions)

26. Would you use an AI system for interview preparation? (Yes/No/Maybe)
27. What features would be most valuable? (Rank top 5)
28. How important is explanation of AI recommendations? (1-5)
29. What concerns do you have about AI career guidance? (Multiple choice)
30. Would you prefer AI feedback or human feedback? (AI/Human/Both)
31. How important is data privacy in career guidance systems? (1-5)
32. What data would you be comfortable sharing? (Multiple choice)
33. How often would you use such a system? (Daily/Weekly/Monthly)
34. Would you pay for AI career guidance services? (Yes/No/Depends)
35. What would make you trust an AI career system? (Open-ended)

Appendix A 2:Survey Instrument - Demographics Section

Survey Objectives

- Understand recruitment challenges and candidate evaluation criteria
- Identify gaps in candidate interview readiness
- Assess value of AI-assisted candidate preparation
- Validate system requirements from employer perspective

Survey Sections & Questions (35 Questions)

Section A: Professional Background (5 Questions)

1. Years of experience in recruitment/HR: 0-2 / 3-5 / 6-10 / 10+
2. Industry sector: Technology / Finance / Healthcare / Other
3. Company size: Startup / SME / Large Enterprise
4. Number of interviews conducted monthly: 1-5 / 6-10 / 11-20 / 20+
5. Primary recruitment focus: Entry-level / Mid-level / Senior

Section B: Candidate Evaluation (10 Questions)

6. What percentage of candidates are well-prepared for interviews?
7. Most common weaknesses observed in candidates? (Rank top 5)
8. How important is communication clarity in hiring decisions? (1-5)
9. Rate importance of technical skills vs soft skills (1-5 each)
10. What interview formats do you use? (Multiple choice)
11. How do you evaluate candidate responses? (Open-ended)
12. What makes a candidate stand out in interviews? (Rank top 3)
13. How often do candidates misalign skills with job requirements? (1-5)
14. Do you provide interview feedback to candidates? (Always/Sometimes/Never)
15. What prevents candidates from performing well? (Multiple choice)

Section C: Recruitment Challenges (8 Questions)

16. Biggest challenges in candidate screening? (Rank top 3)
17. How much time spent per interview on average? (Minutes)
18. Do you use AI tools in recruitment? (Yes/No)
19. If yes, which tools and for what purpose?
20. Satisfaction with current recruitment tools? (1-5)
21. What recruitment process improvements are needed? (Open-ended)
22. How important is candidate self-awareness? (1-5)
23. Would better-prepared candidates improve hiring outcomes? (Yes/No)

Section D: AI System Value (8 Questions)

24. Would you recommend AI interview prep to candidates? (Yes/No)
25. What benefits would AI-prepared candidates bring? (Multiple choice)
26. Concerns about AI-assisted candidate preparation? (Multiple choice)
27. Should AI systems provide transparent explanations? (Yes/No)
28. How important is ethical AI in recruitment? (1-5)

Appendix A 3:Survey Instrument - Current Career & Skills Section

Purpose

To assess current challenges in interview preparation, career decision-making processes, and expectations from AI-driven career guidance systems among students and job seekers.

Section A: Demographic Information

1. Age Group
   - ○ 18-22 years
   - ○ 23-27 years
   - ○ 28-32 years
   - ○ 33+ years

2. Current Status
   - ○ Undergraduate student
   - ○ Postgraduate student
   - ○ Recent graduate (0-2 years)
   - ○ Job seeker (2+ years experience)

3. Field of Study/Work
   - ○ Computer Science/IT
   - ○ Engineering
   - ○ Business/Management
   - ○ Arts/Humanities
   - ○ Science
   - ○ Other

Section B: Interview Preparation Challenges

Rate the following statements (1=Strongly Disagree, 5=Strongly Agree)

4. I find it difficult to effectively communicate my skills during interviews
   ○ 1 ○ 2 ○ 3 ○ 4 ○ 5

5. I struggle to structure my responses in a clear and professional manner
   ○ 1 ○ 2 ○ 3 ○ 4 ○ 5

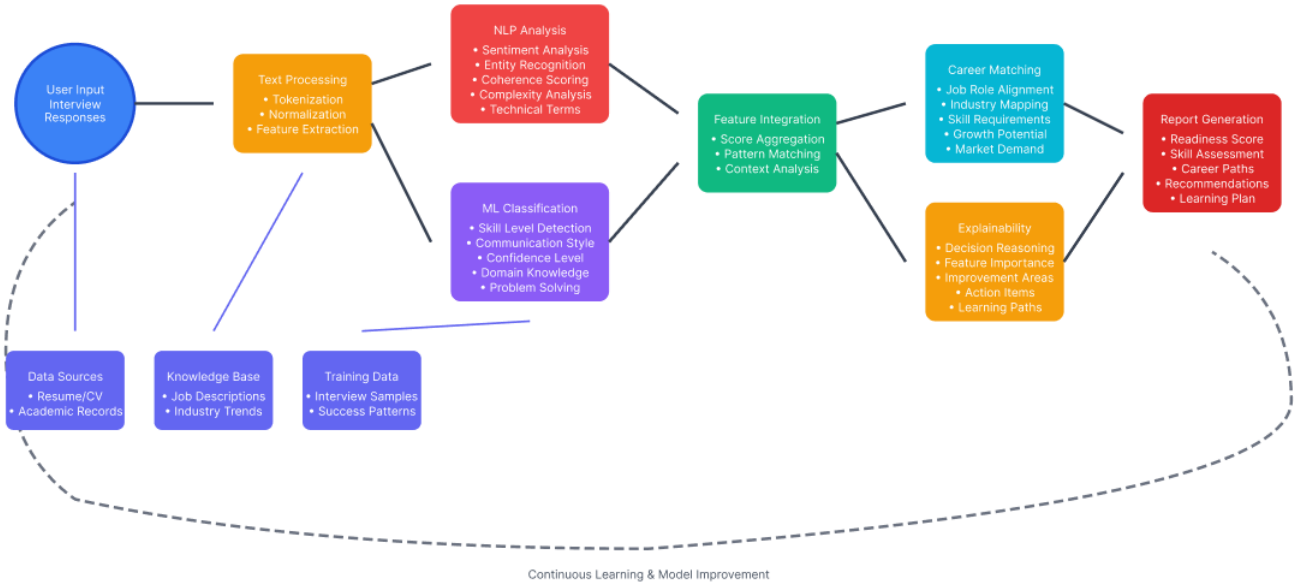
6. I lack confidence when discussing my professional background
   ○ 1 ○ 2 ○ 3 ○ 4 ○ 5

7. I am uncertain about which career paths align with my skills
   ○ 1 ○ 2 ○ 3 ○ 4 ○ 5

8. What are your biggest challenges when preparing for job interviews? (Open-ended)
   Please describe your main challenges...

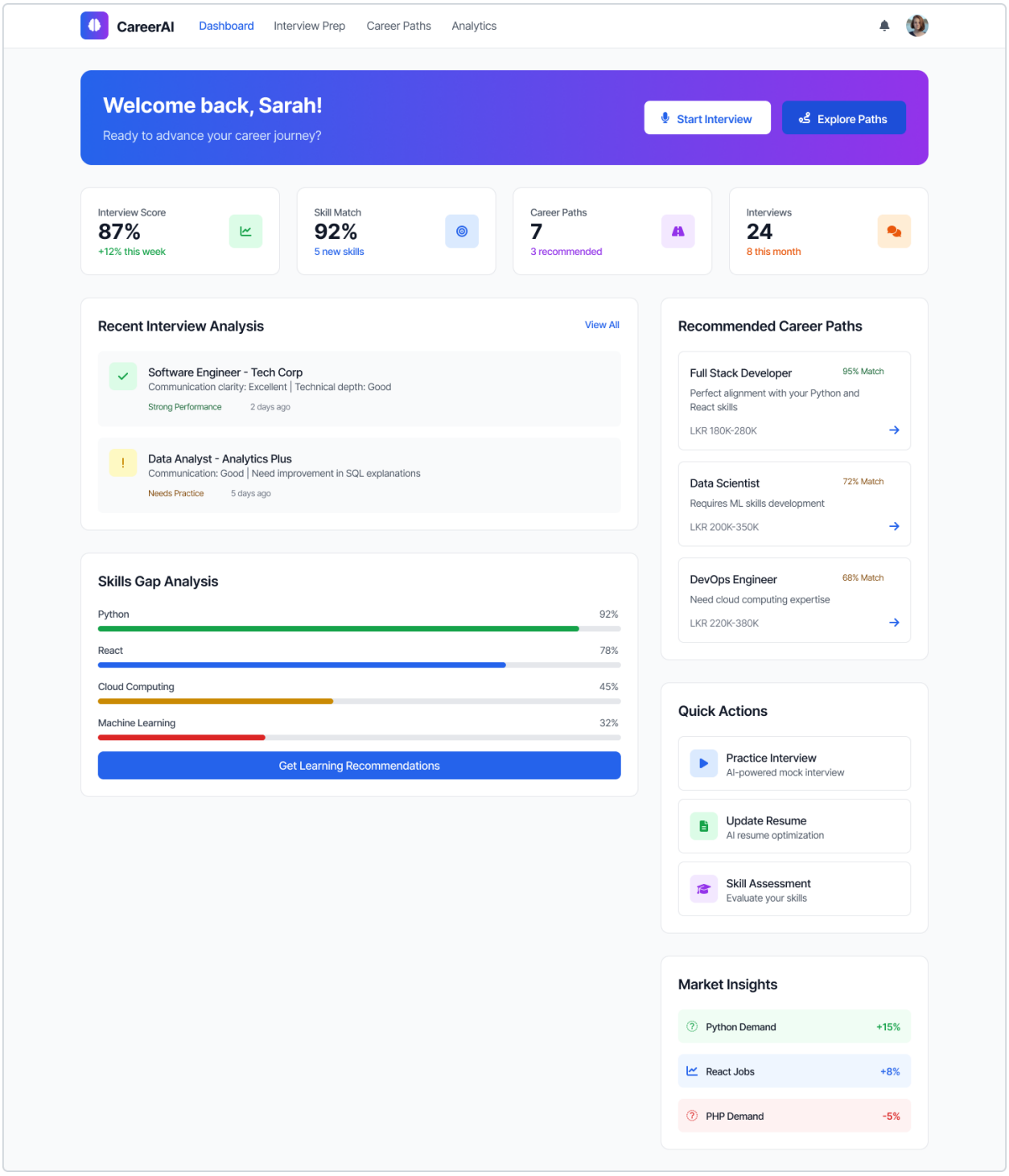
Appendix A 4:Survey Instrument - Career Guidance & AI Preferences Section

11.2 Detailed diagrams

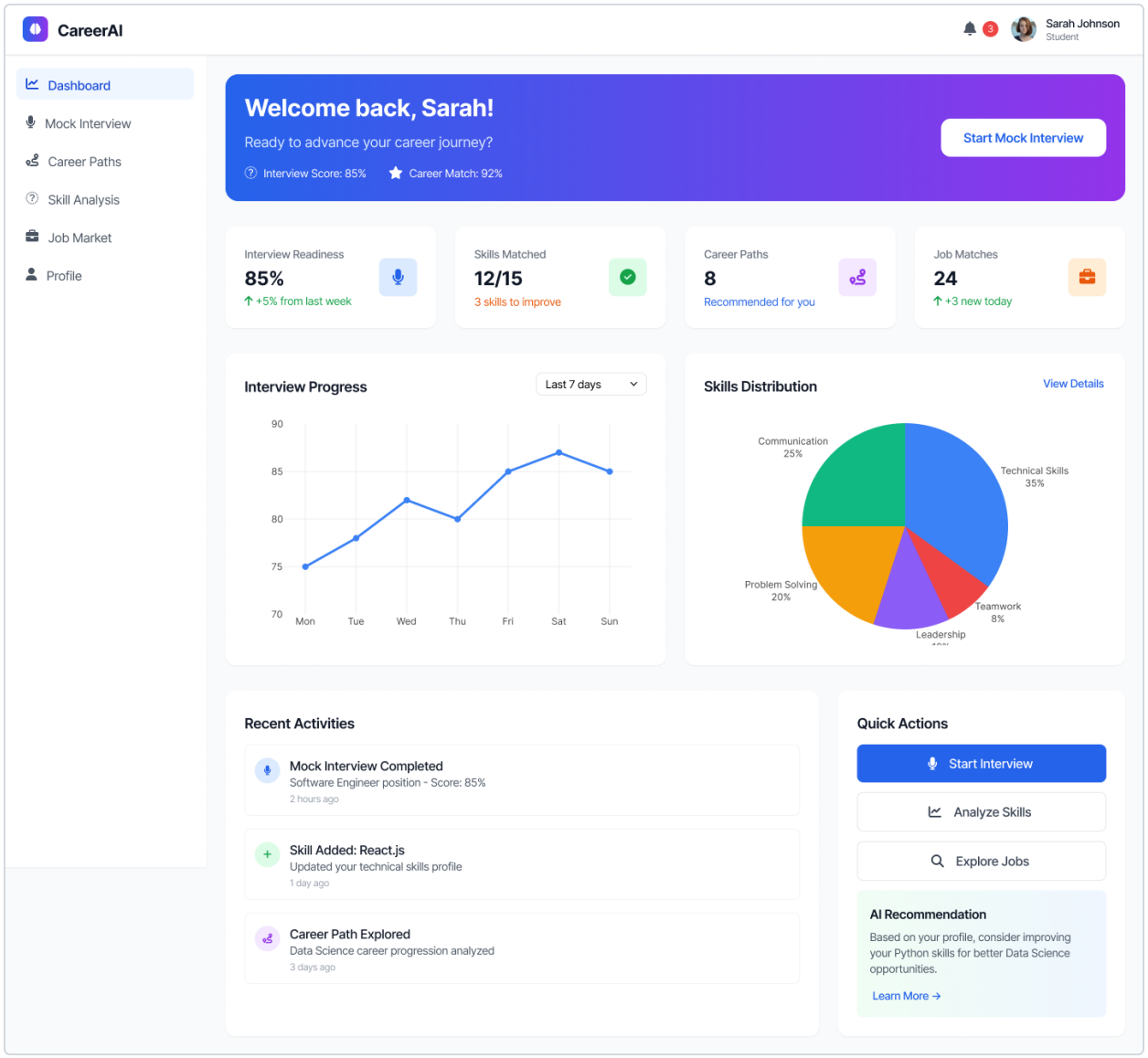


Appendix B 1:Detailed Diagram - System Component Interaction

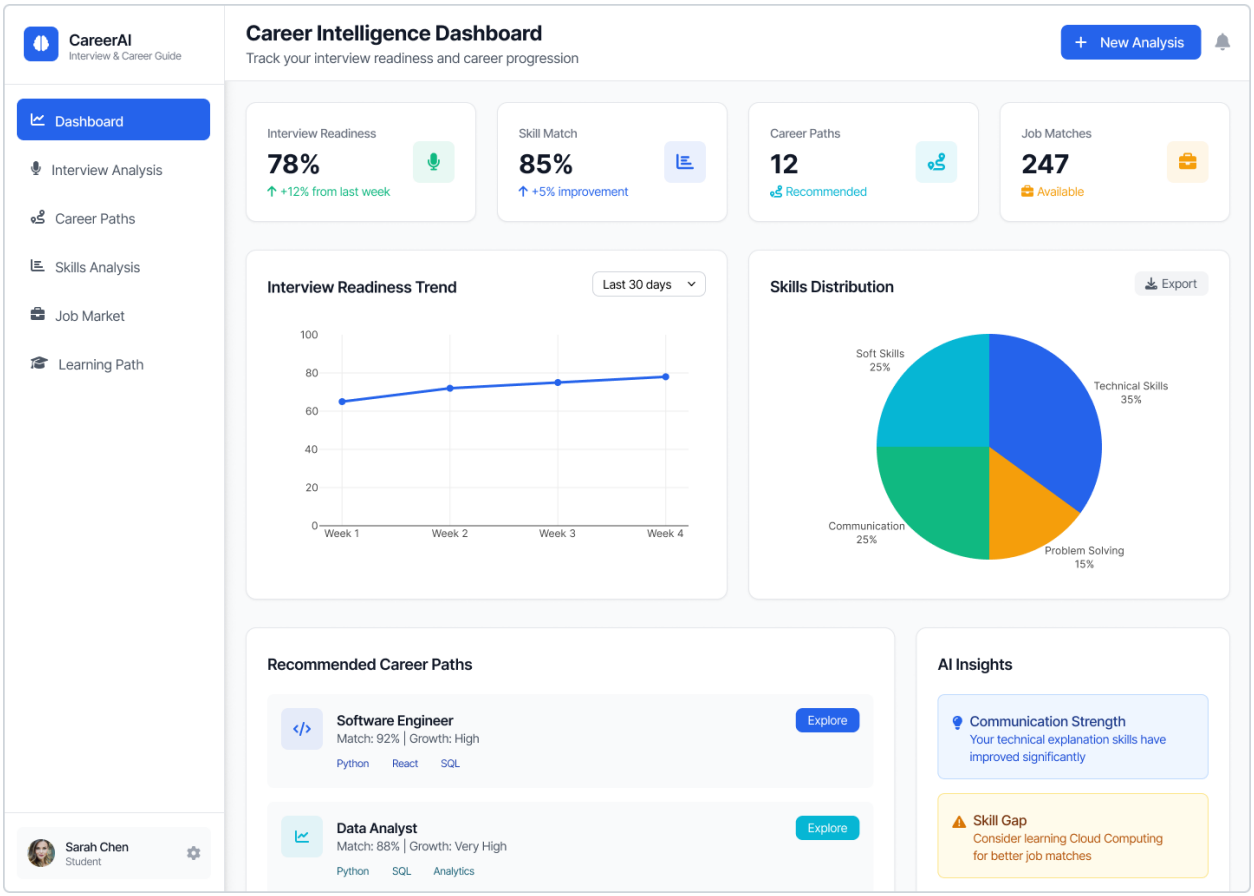
11.3 Preliminary UI mockups



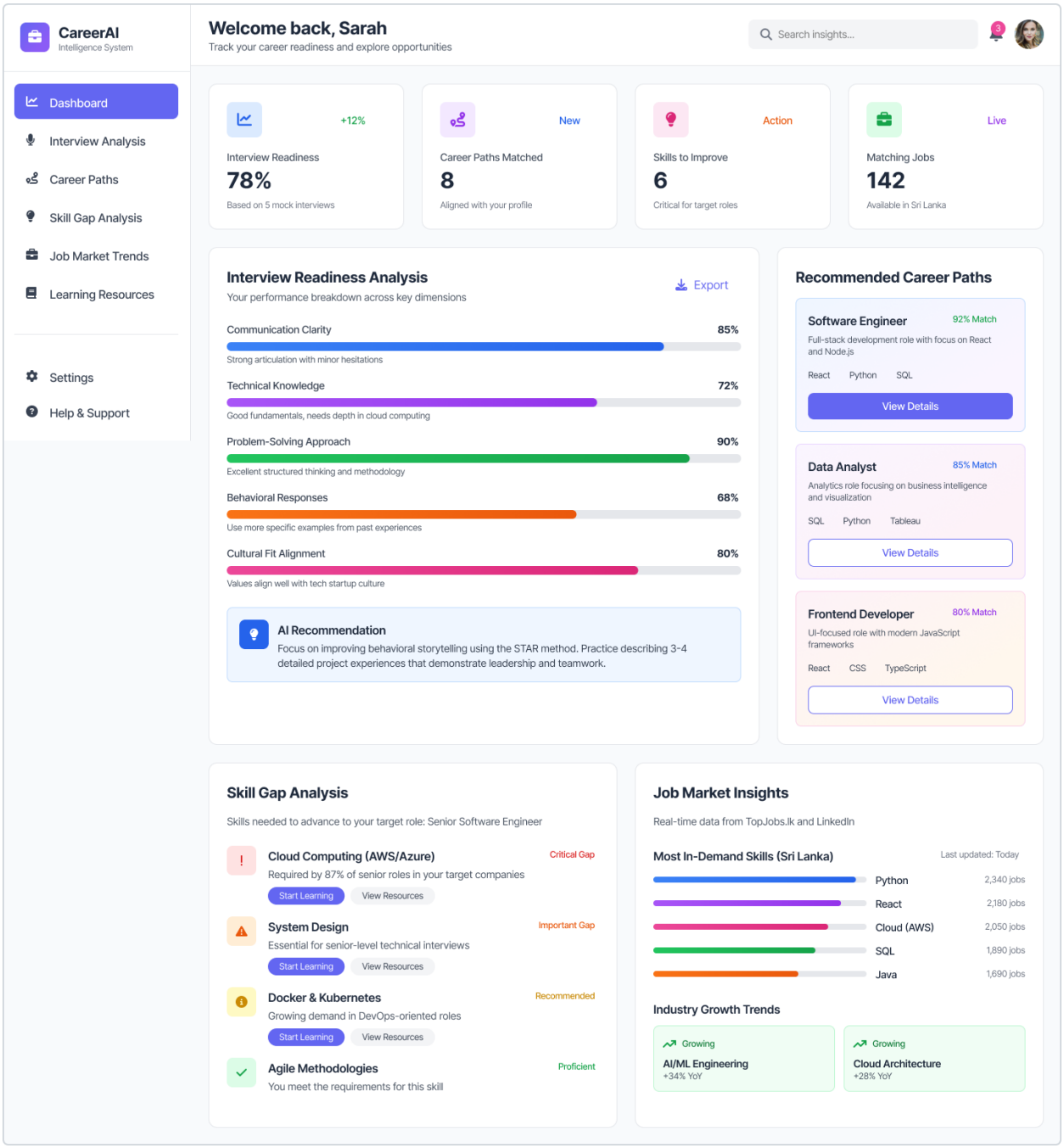
Appendix C 1:UI Mockup - CareerAI Dashboard



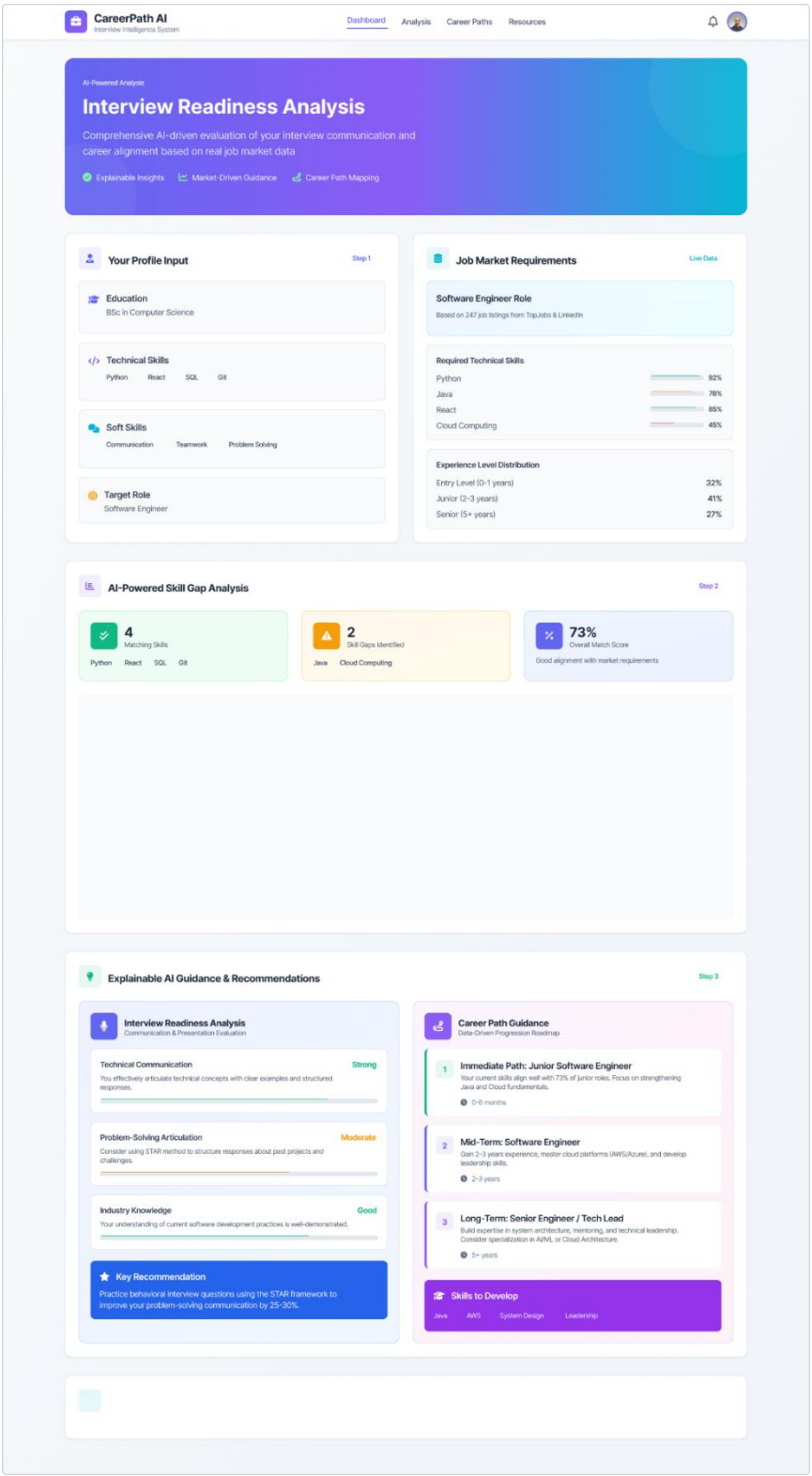
Appendix C 2:UI Mockup - Career Intelligence Dashboard



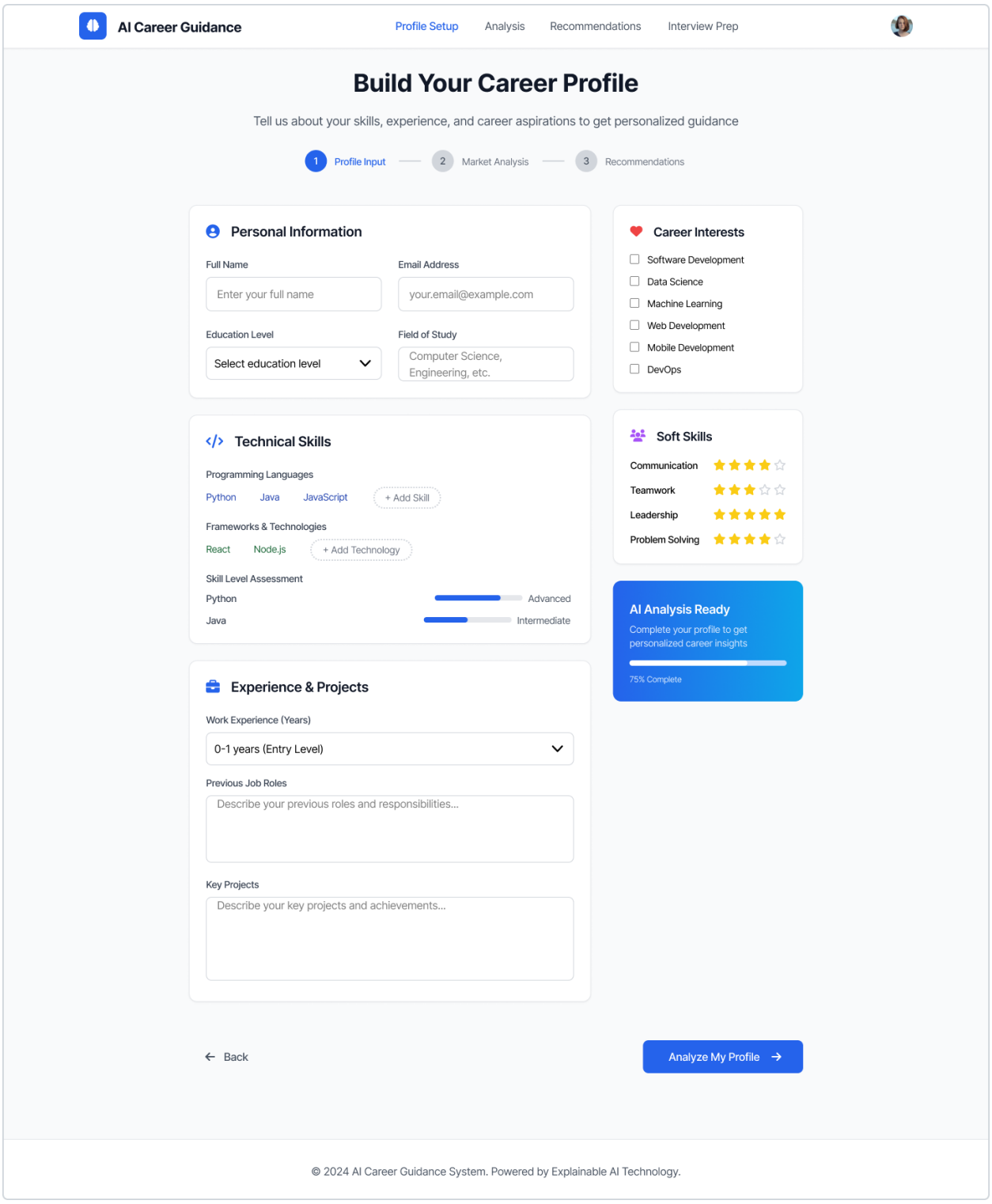
Appendix C 3:UI Mockup - CareerAI Dashboard



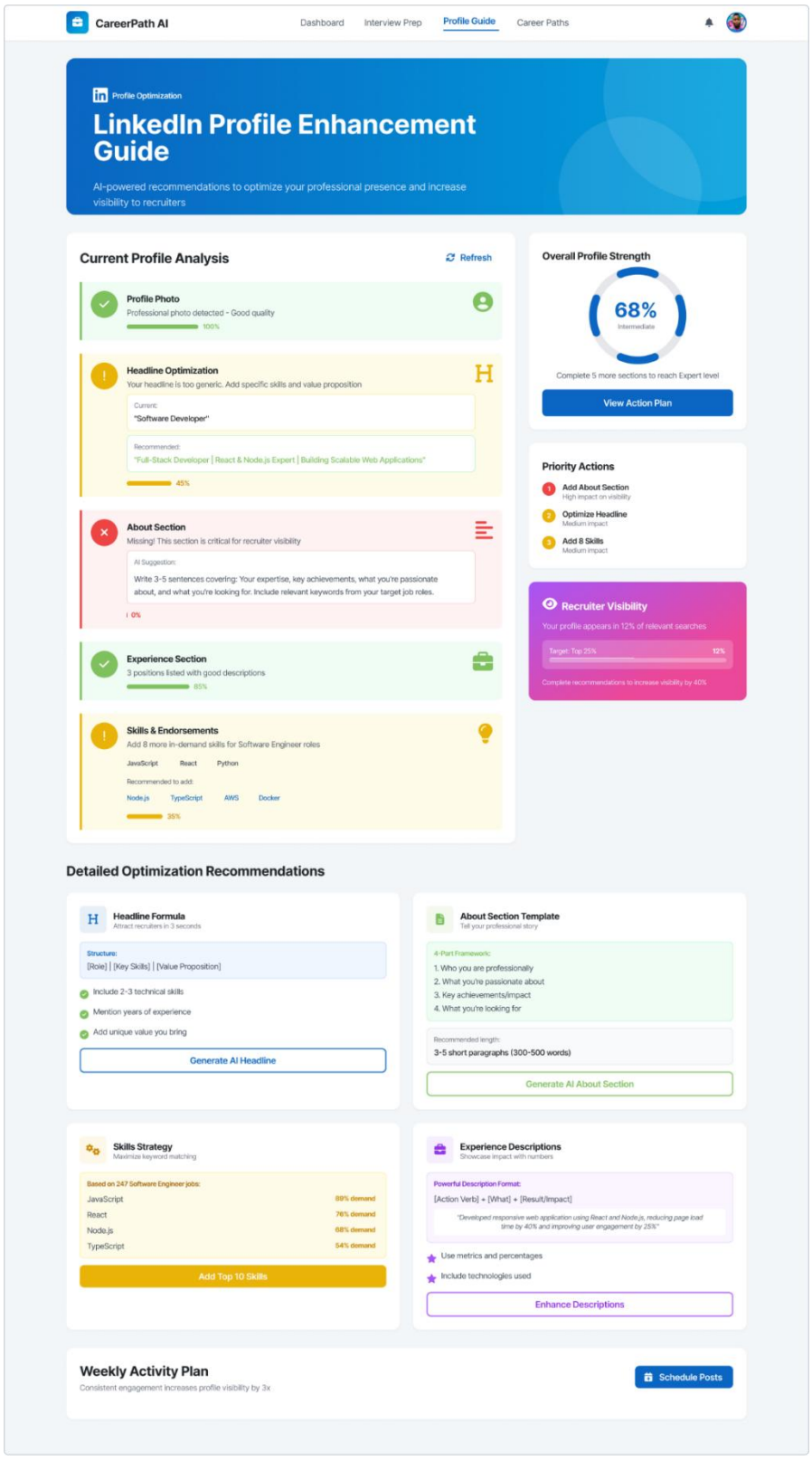
Appendix C 4: UI Mockup - CareerAI Welcome Dashboard



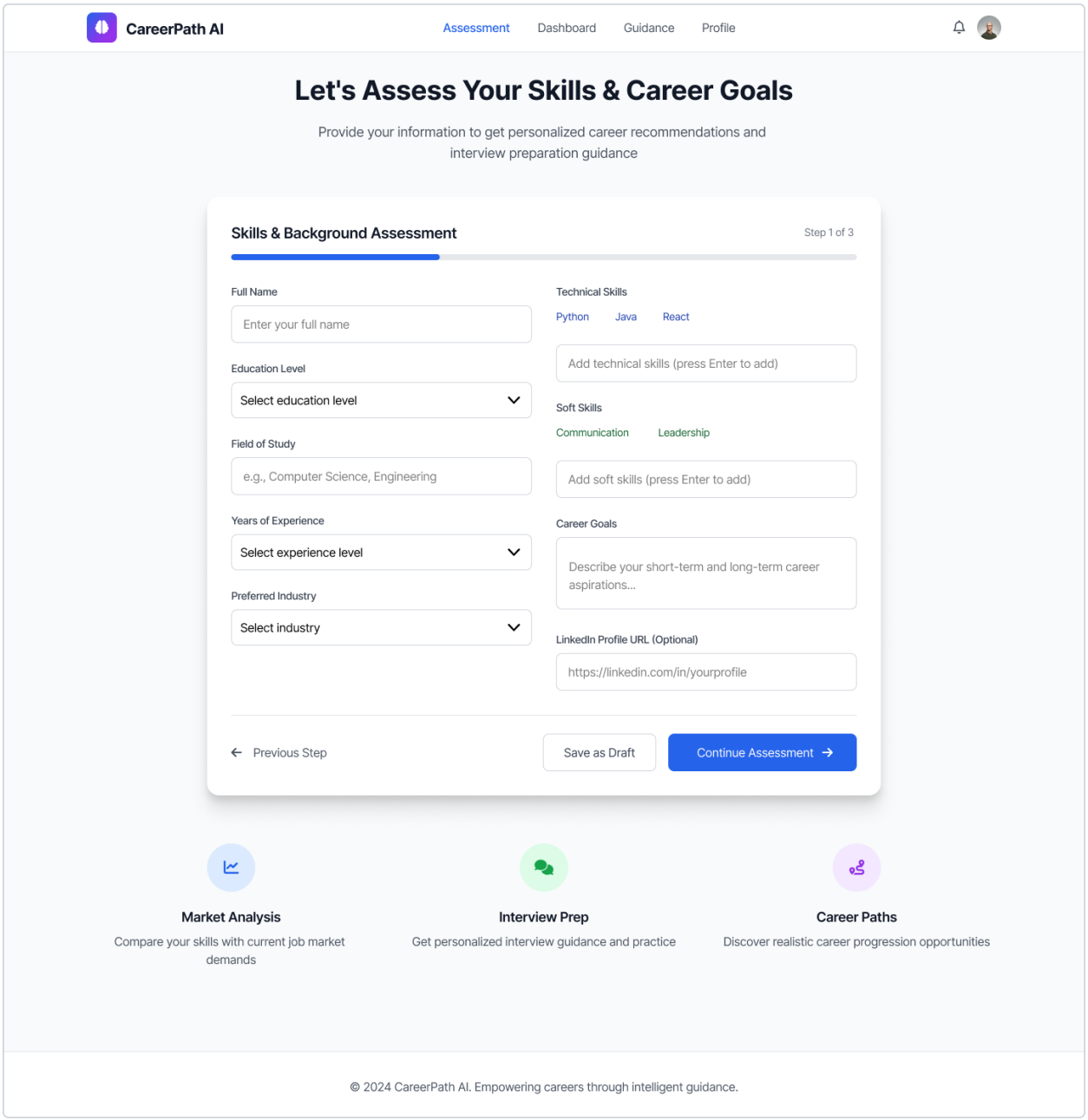
Appendix C 5: UI Mockup - CareerPath AI - Interview Readiness Analysis Results



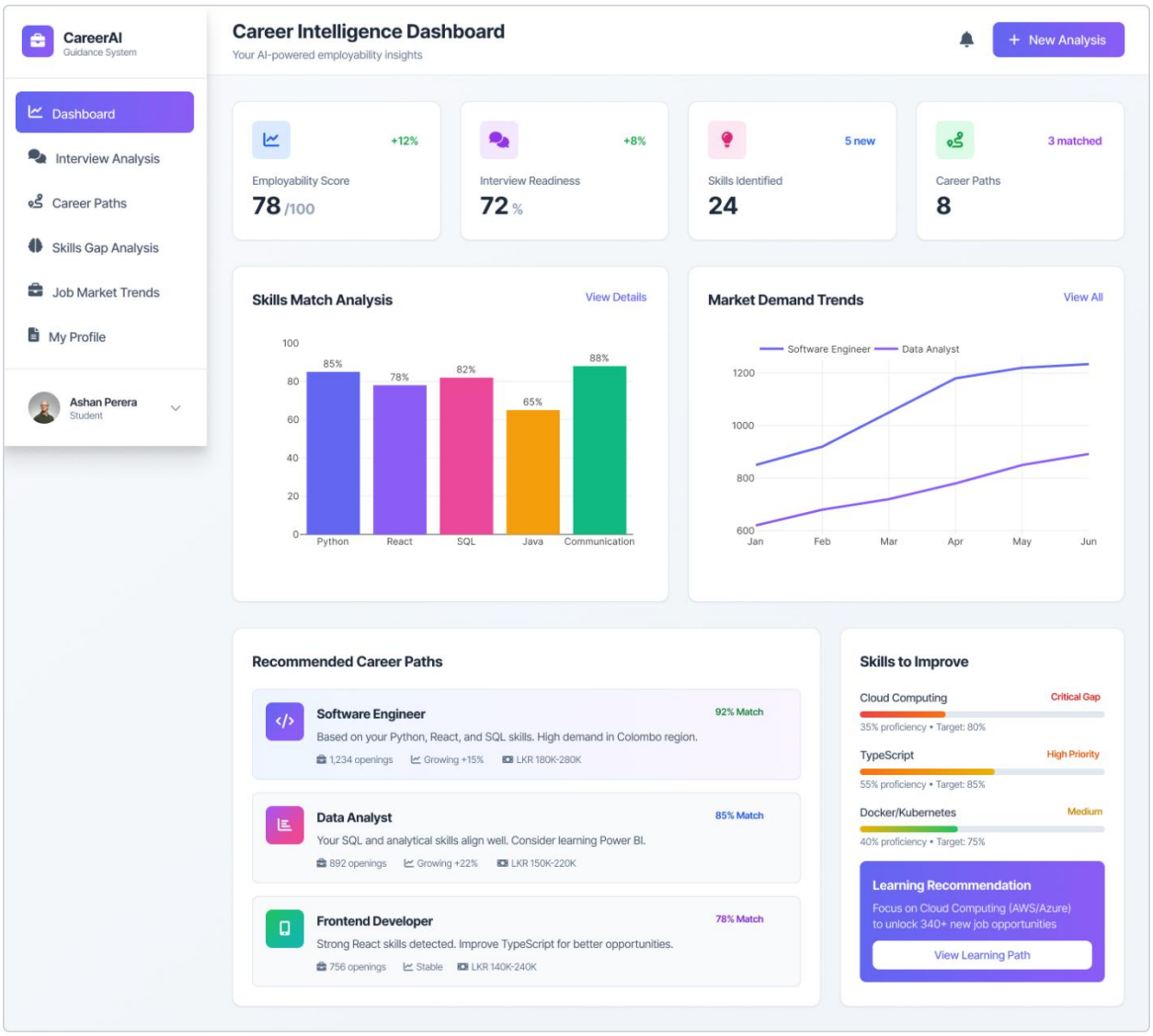
Appendix C 6: UI Mockup - Career Profile Builder



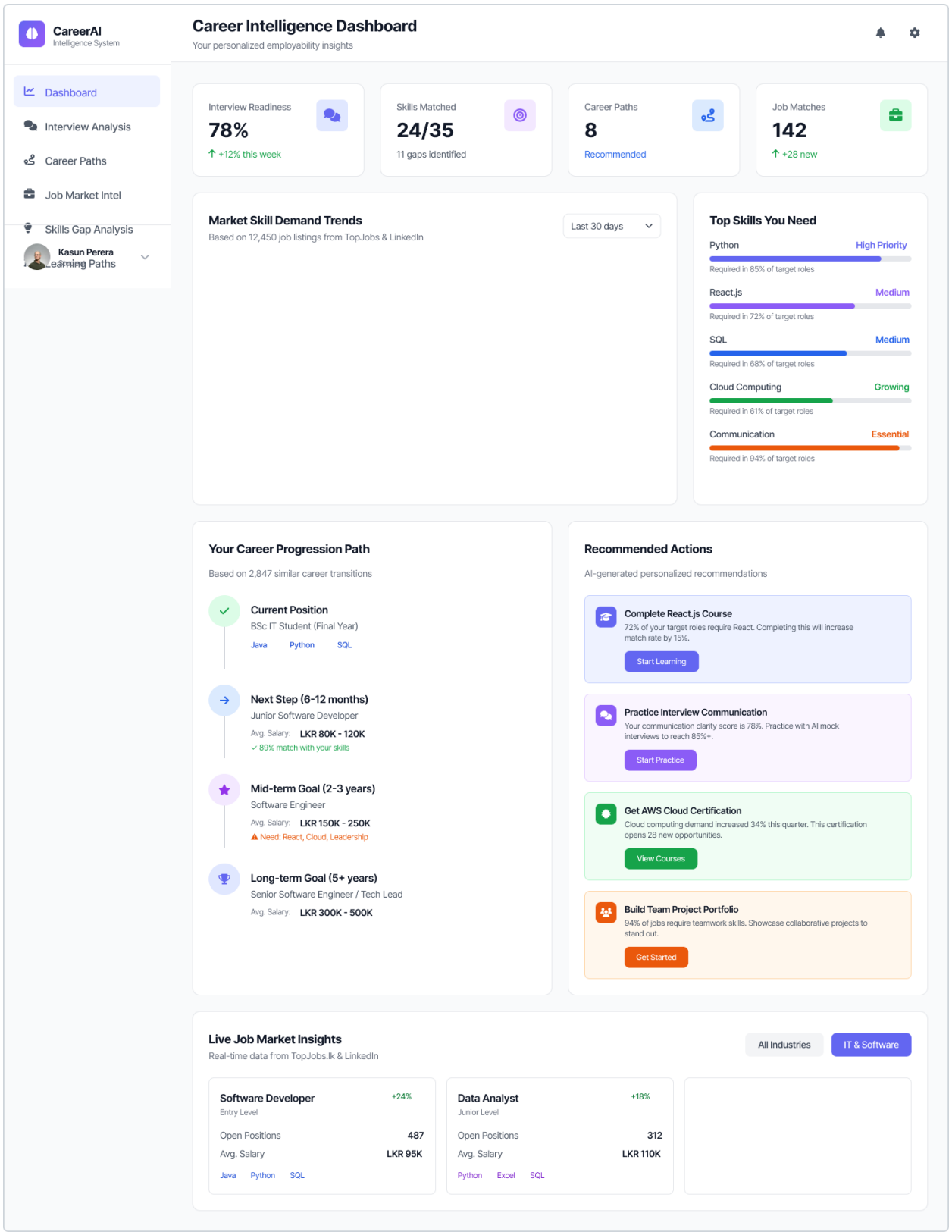
Appendix C 7:UI Mockup - CareerPath AI - LinkedIn Profile Enhancement Guide



Appendix C 8:UI Mockup - CareerPath AI - Skills & Career Goals Assessment Form



Appendix C 9:UI Mockup - Career Intelligence Dashboard (Employability Insights)



Appendix C 10: UI Mockup - Career Intelligence Dashboard (Skills & Paths Analysis)

11.4 Risk analysis

Table 7:Risk analysis table

Risk Category	**Description**	**Mitigation Strategy**
Data Privacy Risk	Personal information such as CVs or interview transcripts may contain sensitive data.	Anonymize personal data and store information securely.
AI Bias Risk	AI models may produce biased results if training data is unbalanced.	Use diverse datasets and regularly evaluate model fairness.
Misinterpretation Risk	Users may misunderstand system feedback as a final hiring decision.	Clearly explain that the system is a decision-support tool, not an automated recruiter.
Model Accuracy Risk	NLP models may misinterpret certain responses or language patterns.	Perform model testing and continuous validation using evaluation metrics.
System Security Risk	Unauthorized access to stored candidate data.	Use secure authentication and encrypted data storage.
Ethical Misuse Risk	The system could be used to unfairly judge candidates.	Provide transparent explanations and ensure human oversight.